The Land Repair Helix

Site plan: Riverside land

Site name: _____ Date: _____ Prepared by: _____

Bare soil over winter → Keep it green (Soil)

Bare soil erodes in heavy rain and carries sediment and phosphorus into rivers and loughs.

- ☐ £ Leave stubbles, or sow a winter cover crop
- ☐ £ Keep grass margins at field edges and gateways
- ☐ ££ Plant hedges or grass strips across slopes to slow runoff

Flooding and fast runoff → Slow the flow (Water)

Hard surfaces and drained land send water downstream fast, causing floods and washing pollution off the land.

- ☐ £ Break up hard surfaces with gravel, grass-crete or permeable paving
- ☐ ££ Dig swales, rain gardens and small ponds to hold water back
- ☐ ££ Install leaky dams of logs or brash in ditches and small streams
- ☐ £££ Add sustainable drainage (SuDS) when sites are redeveloped

NI: Work in or near a watercourse may need consent from DfI Rivers. Check before building dams or culverts.

Blue-green algae (cyanobacteria) blooms → Starve the bloom of phosphorus (Water)

Too much phosphorus and nitrogen, plus warm still water, feeds toxic cyanobacteria. Lough Neagh is the clearest NI example.

- ☐ £ Test soil phosphorus and stop applying where the index is already high
- ☐ £ Fence livestock out of streams and provide drinking troughs
- ☐ £ Get septic tanks emptied and checked
- ☐ ££ Leave wide, ungrazed buffer strips of grass and trees along watercourses
- ☐ £££ Build constructed wetlands or reedbeds to catch nutrients from yards and drains

NI: Keep slurry and fertiliser off land next to waterways, and inside the buffer distances, especially in winter and before forecast rain.

Take care: Don't touch bloom scums, and keep dogs out of the water. Cyanobacteria can be toxic.

Diesel, oil or paraffin spilled on land → Contain, soak up, dispose properly (Spills)

A few litres of oil can pollute a large area of soil, a drain or a stream.

- ☐ £ Stop the leak and block nearby drains with covers or sandbags
- ☐ £ Soak it up with sawdust, wood shavings, straw, coir, cork granules or wool
- ☐ £ Clay granules (cat litter) work well on hard surfaces
- ☐ £ Keep a ready-made spill kit beside every fuel store and generator
- ☐ ££ Dig out badly soaked soil, then let microbes treat it in a biopile

NI: If oil reaches a drain, watercourse or groundwater, ring the NIEA Incident Hotline on 0800 80 70 60. Used sorbents are hazardous waste and must go via a registered carrier.
Take care: Don't use peat, which destroys bogs, or washing-up liquid and detergents, which spread the oil into the water.

Trees in the wrong place → Right tree, right place (Air and carbon)

Trees planted on species-rich grassland, peat or nesting habitat can do more harm than good.

- ☐ £ Plant native broadleaves along watercourses for shade; cooler water helps reduce algal blooms
- ☐ £ Plant hedgerow trees and shelterbelts rather than blocks on good habitat
- ☐ ££ Let woodland regenerate naturally where there are seed sources

NI: Check for priority habitat and protected sites before planting.

Invasive plants spreading → Catch them early, treat properly (Rewilding)

Japanese knotweed, Himalayan balsam and rhododendron crowd out native plants, damage structures and spread along rivers and roads.

- ☐ £ Learn to recognise them and map where they are
- ☐ £ Pull Himalayan balsam before it sets seed (June to July), and repeat each year
- ☐ ££ Treat knotweed with a specialist herbicide programme over several seasons
- ☐ £££ Excavate knotweed where development needs it, under a management plan

NI: It's an offence in NI to plant certain invasive species, or cause them to grow, in the wild. Knotweed waste is controlled waste and must go to an authorised site.
Take care: Never strim or flail knotweed: tiny fragments regrow.

Litter and fly-tipping → Design it out, report it fast (Rewilding)

Litter and dumped waste blight places, harm wildlife and signal neglect, which draws more dumping.

- ☐ £ Report fly-tipping to the council quickly; most NI councils take reports online
- ☐ £ Organise or support community clean-ups, such as Live Here Love Here
- ☐ ££ Fix lighting, fencing and sightlines at dumping hotspots
- ☐ ££ Use a registered waste carrier and keep transfer notes: your duty of care doesn't end at the gate

NI: Councils enforce litter and fly-tipping law, and NIEA deals with larger illegal waste sites.

Take care: Don't move dumped waste that may be hazardous, such as asbestos or drums. Report it instead.

Important. For contaminated land, forever chemicals (PFAS) or any spill reaching a drain, watercourse or groundwater, get specialist advice and involve the regulator. In NI, report pollution to the NIEA Incident Hotline on **0800 80 70 60** (24 hours). Contaminated-land remediation normally goes through planning, using the Land Contamination: Risk Management (LCRM) framework.

Sources. DAERA and CAFRE guidance on the Nutrients Action Programme and the hedge-cutting closed period (1 March to 31 August); NIEA incident reporting (nidirect); LCRM (Environment Agency, used by NIEA); UK and EU work on PFAS (Defra PFAS Plan 2026, ECHA); the All-Ireland Pollinator Plan; the "Don't Mow, Let It Grow" initiative run by NI councils and partners.

Position at 16 October 2026. For information, not legal or engineering advice. Robert J. Park · linkedin.com/in/robertparkuk · WhatsApp @robertjpark